MATH 1342 – Elementary Statistics – Week 2: Sampling, Experiments and Observational Studies

Sep 1 – 3 | 1.2 Sampling from a Population • 1.3 Experiments and Observational Studies | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Distinguish a population from a sample and describe statistical inference. (CLO 1) **2.** Identify sampling bias in a described study, classify the sampling method (random, convenience, volunteer), and judge whether the results can be generalized. (CLO 1) **3.** Recognize bias from question wording, question context, and inaccurate responses. (CLO 1)	e-Text: 1.2 Sampling from a Population Interactive Lecture: 1.2 Sampling from a Population Slides: 1.2 Sampling from a Population (includes sampling video)	Participation: Variables in Statistics (Kahoot!) – Tue: review of Week 1 Participation: Sampling – spot the bias in sample scenarios Participation: Sampling Methods (Kahoot!) – Thu: sampling review	Online Homework: 1.2 Sampling from a Population (due Sun Sep 6, 11:59 PM)	WileyPLUS StatKey Blackboard Kahoot! Zoom (live class)
4. Classify a study as observational or experimental, and identify a plausible confounding variable. (CLO 1) **5.** Explain how random assignment, a control group, and blinding support cause-and-effect conclusions, and distinguish random sampling (generalize) from random assignment (causation). (CLO 1)	e-Text: 1.3 Experiments and Observational Studies Interactive Lecture: 1.3 Experiments and Observational Studies Slides: 1.3 Experiments and Observational Studies (includes TED-Ed placebo video)	Participation: Experiments and Observational Studies – classify studies; confounding; random sampling vs. random assignment	Online Homework: 1.3 Experiments and Observational Studies (due Sun Sep 6, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom (live class)
Items that cover objectives from more than one row above (objectives in parentheses)			Take-Home Quiz 1 (due Tue Sep 8, 11:59 PM) – covers Intro, 1.1–1.3 (all objectives)	Assignment Tool in Blackboard

CLO 1 Explain the use of data collection and statistics as tools to reach reasonable conclusions.